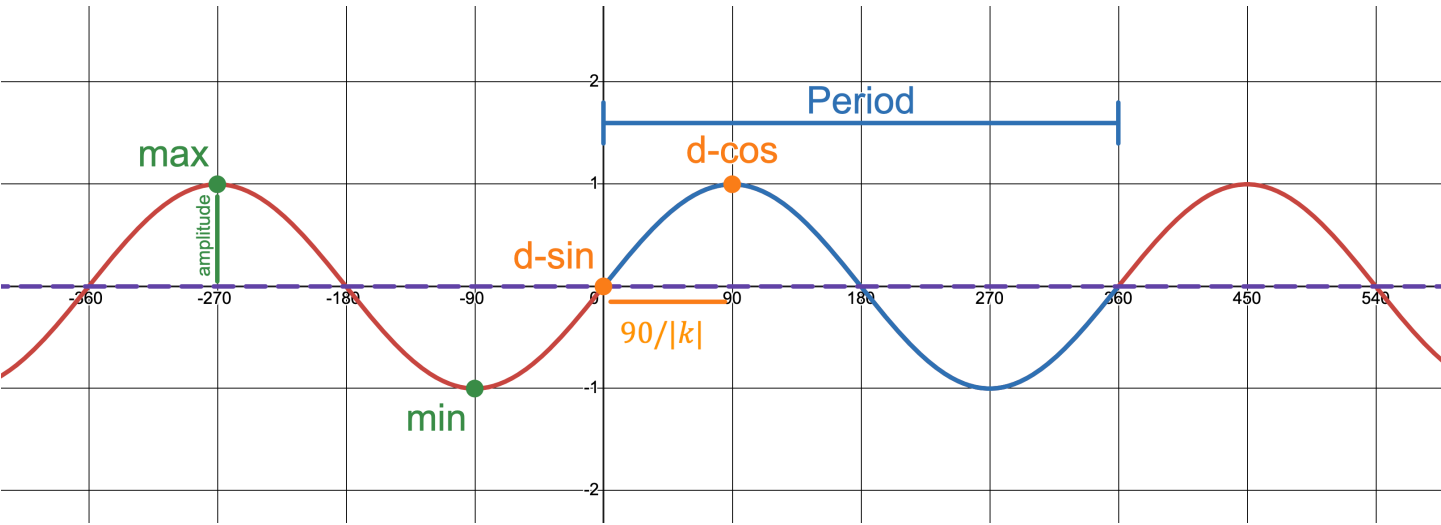


Section 1: How to Determine the Equation of a Sine or Cosine Function Given its Graph

To determine the equation of a transformed sine or cosine function, you must use the graph to determine the value of the parameters $a, k, d,$ and c to then write the transformed equation in the format:

$y = a \sin[k(x - d)] + c$ OR $y = a \cos[k(x - d)] + c$

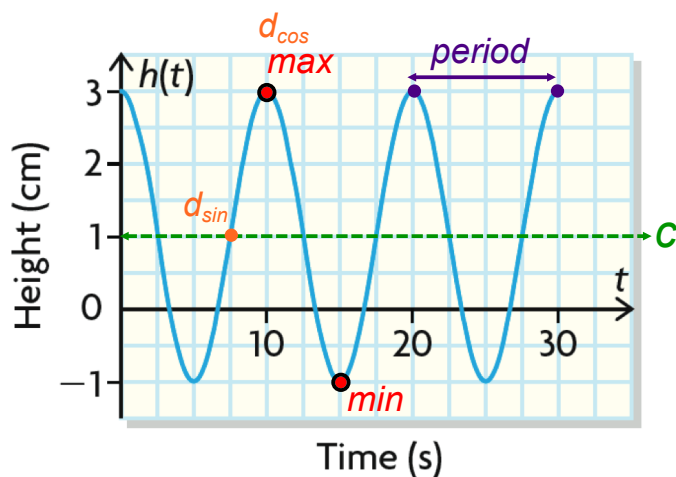
a	k	d	c
Find the amplitude of the function: $a = \frac{y_{max} - y_{min}}{2}$	Find the period (in radians) of the function using a starting point and ending point of a full cycle. $k = \frac{360}{period}$	for sin x: x-coordinate of a rising mid-line. for cos x: x-coordinate of a maximum point. $d_{sin} = d_{cos} - \frac{90}{ k }$ $d_{cos} = d_{sin} + \frac{90}{ k }$	Find the vertical shift $c = max - amplitude$ OR $c = \frac{max + min}{2}$ (this finds the ‘middle’ of the function)



Section 2: Determining the Equation of a Sinusoidal Function Given its Graph

Example 1: For each of the following graphs, determine the equation of a sine and cosine function that represents each graph:

a)



$$a = \frac{\max - \min}{2} = \frac{3 - (-1)}{2} = 2$$

$$k = \frac{360}{\text{period}} = \frac{360}{30 - 20} = \frac{360}{10} = 36$$

$$c = \max - |a| = 3 - 2 = 1$$

$$d_{\sin} = 7.5$$

$$d_{\cos} = 10$$

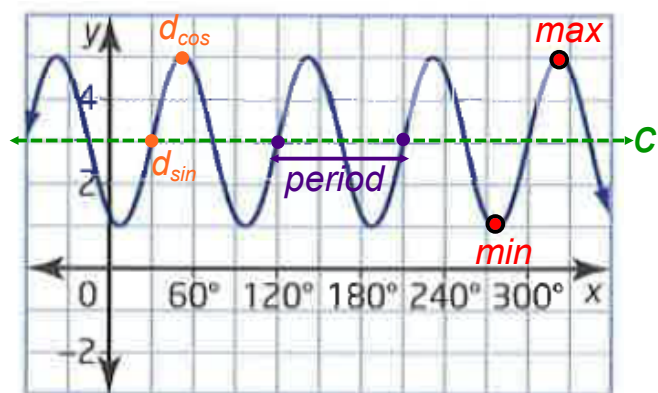
$d_{\sin} \rightarrow$ look for x -value of closest rising midline

$d_{\cos} \rightarrow$ look for x -value of closest maximum

$$y = 2 \cos[36(x - 10)] + 1$$

$$y = 2 \sin[36(x - 7.5)] + 1$$

b)



$$a = \frac{\max - \min}{2} = \frac{5 - 1}{2} = 2$$

$$k = \frac{360}{\text{period}} = \frac{360}{210 - 120} = \frac{360}{90} = 4$$

$$c = \max - |a| = 5 - 2 = 3$$

$$d_{\sin} = 30$$

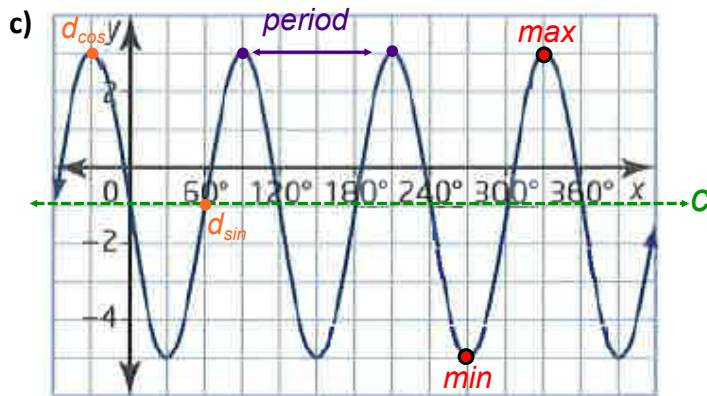
$$d_{\cos} = d_{\sin} + \frac{90}{|k|} = 30 + \frac{90}{4} = 52.5$$

$$y = 2 \cos[4(x - 52.5)] + 3$$

$$y = 2 \sin[4(x - 30)] + 3$$

Note: The x - value of the maximum point was not obvious from the graph. You need to know that maximum points are always $\frac{90}{|k|}$ to the right of the rising midline point. Also, if you knew where the maximum point was, the rising midline point would be $\frac{90}{|k|}$ to the left of the max.

$$d_{\cos} = d_{\sin} + \frac{90}{|k|} \quad \text{OR} \quad d_{\sin} = d_{\cos} - \frac{90}{|k|}$$



$$a = \frac{\text{max} - \text{min}}{2} = \frac{3 - (-5)}{2} = 4$$

$$k = \frac{360}{\text{period}} = \frac{360}{210 - 90} = \frac{360}{120} = 3$$

$$c = \text{max} - |a| = 3 - 4 = -1$$

$$d_{\sin} = 60$$

$$d_{\cos} = -30$$

$$y = 4 \cos[3(x + 30)] - 1$$

$$y = 4 \sin[3(x - 60)] - 1$$

Example 2: A sinusoidal function has an amplitude of 3 units, a period of 180 degrees and a max point at (0, 5). Represent the function with an equation in two different ways.

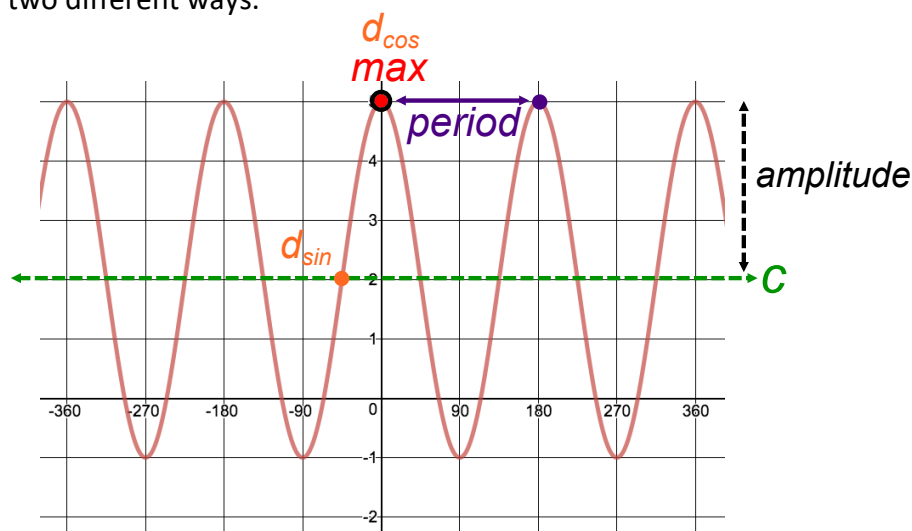
$$a = 3$$

$$k = \frac{360}{\text{period}} = \frac{360}{180} = 2$$

$$c = \text{max} - |a| = 5 - 3 = 2$$

$$d_{\cos} = 0$$

$$d_{\sin} = d_{\cos} - \frac{90}{|k|} = 0 - \frac{90}{2} = -45$$



$$y = 3 \cos(2x) + 2$$

$$y = 3 \sin[2(x + 45)] + 2$$

Example 3: A sinusoidal function has an amplitude of 5 units, a period of 120 degrees and a maximum at (0, 3). Represent the function with an equation in two different ways.

$$a = 5$$

$$k = \frac{360}{\text{period}} = \frac{360}{120} = 3$$

$$c = \text{max} - |a| = 3 - 5 = -2$$

$$d_{\cos} = 0$$

$$d_{\sin} = d_{\cos} - \frac{90}{|k|} = 0 - \frac{90}{3} = -30$$

$$y = 5 \cos(3x) - 2$$

$$y = 5 \sin[3(x + 30)] - 2$$

